

Solving the Time-Dependent Schrödinger Equation

via the Crank–Nicolson Method: Stability, Unitarity & Efficient Linear Algebra

A complete mathematical derivation from operator formalism to tridiagonal linear systems and the Thomas algorithm

1. The Physical Problem

The **Time-Dependent Schrödinger Equation** (TDSE) governs quantum state evolution:

$$i\hbar \frac{\partial \Psi(x,t)}{\partial t} = \hat{H} \Psi(x,t) = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \Psi(x,t) \quad (1)$$

Formal solution — time-evolution operator:

$$\Psi(x, t+\Delta t) = \underbrace{e^{-i\hat{H}\Delta t/\hbar}}_{\hat{U}(\Delta t)} \Psi(x, t)$$

Unitarity of $\hat{U}(\Delta t)$ is essential for probability conservation. This is guaranteed if $\hat{H}$ is Hermitian.

Unitarity: the Core Requirement	$\hat{H} = \hat{H}^\dagger$ (Hermitian) $\Rightarrow \hat{U}^\dagger \hat{U} = e^{+i\hat{H}\Delta t/\hbar} e^{-i\hat{H}\Delta t/\hbar} = \hat{I}$ This enforces probability conservation : $\int_{-\infty}^{+\infty} \Psi(x, t) ^2 dx = 1 \quad \forall t$ <i>Any valid numerical scheme must discretely replicate this.</i>
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The computational challenge: approximate $e^{-i\hat{H}\Delta t/\hbar}$ on a finite spatial grid while keeping the discrete propagator *exactly* unitary.

2. Space–Time Discretization

Introduce a uniform grid with $N + 1$ spatial and M temporal points:

$$x_j = j \Delta x, \quad j = 0, \dots, N; \quad t_n = n \Delta t, \quad n = 0, 1, \dots$$

Central finite difference (2nd-order, $\mathcal{O}(\Delta x^2)$):

$$\left. \frac{\partial^2 \Psi}{\partial x^2} \right|_{x_j} \approx \frac{\Psi_{j+1}^n - 2\Psi_j^n + \Psi_{j-1}^n}{(\Delta x)^2} \quad (2)$$

Derived by Taylor expansion: $\Psi_{j\pm 1} = \Psi_j \pm \Delta x \Psi' + \frac{(\Delta x)^2}{2} \Psi'' \pm \frac{(\Delta x)^3}{6} \Psi''' + \mathcal{O}(\Delta x^4)$

The **discrete Hamiltonian** H_{FD} acts on the grid vector $\Psi^n = [\Psi_1^n, \dots, \Psi_N^n]^\top$:

$$(H_{FD} \Psi^n)_j = -\beta \Psi_{j-1}^n + (2\beta + V_j) \Psi_j^n - \beta \Psi_{j+1}^n, \quad \beta \equiv \frac{\hbar^2}{2m(\Delta x)^2}$$

Boundary conditions are implemented by modifying the first and last rows of H_{FD} .

Boundary Conditions	Dirichlet (infinite well): $\Psi_0^n = \Psi_N^n = 0$ Periodic (ring/lattice): $\Psi_0^n = \Psi_N^n, \Psi_{-1}^n = \Psi_{N-1}^n$ Absorbing (open boundary): Add $-i\Gamma(x)$ to $V(x)$ near edge
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3. Why Euler Schemes Fail: Von Neumann Analysis

Stability test: substitute $\Psi_j^n = g^n e^{ikj\Delta x}$ (Fourier mode) into each scheme and compute amplification factor g .

Forward Euler (FTCS) $\Psi^{n+1} = (I - \frac{i\Delta t}{\hbar} H) \Psi^n$ $ g = \sqrt{1 + (\frac{\Delta t \lambda}{\hbar})^2} > 1$ Unconditionally unstable. Norm grows $\rightarrow$ blows up.	Backward Euler (BTCS) $\Psi^{n+1} = (I + \frac{i\Delta t}{\hbar} H)^{-1} \Psi^n$ $ g = \frac{1}{\sqrt{1 + (\Delta t \lambda / \hbar)^2}} < 1$ Stable, but artificially dissipative. Norm decays $\Rightarrow$ violates QM.	Crank–Nicolson Average of FTCS & BTCS $ g = \left \frac{1 - i\Delta t \lambda / (2\hbar)}{1 + i\Delta t \lambda / (2\hbar)} \right = 1$ Unconditionally stable. Exact norm conservation.
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Here λ denotes any eigenvalue of H_{FD} ; since H_{FD} is real-symmetric all eigenvalues are real, so $|g| = 1$ holds for *every* mode simultaneously — unconditional stability.

4. Crank–Nicolson: Derivation & Cayley Transform

Average the PDE right-hand side at levels n and $n+1$:

$$i\hbar \frac{\Psi_j^{n+1} - \Psi_j^n}{\Delta t} = \frac{1}{2} (H_{FD} \Psi_j^{n+1} + H_{FD} \Psi_j^n) \quad (3)$$

Collect unknowns (Ψ^{n+1}) on the left, with $\gamma \equiv \frac{\Delta t}{2\hbar}$:

$$\underbrace{(I + i\gamma H)}_A \Psi^{n+1} = \underbrace{(I - i\gamma H)}_B \Psi^n \quad (4)$$

The matrix A is tridiagonal and invertible. The matrix B is also tridiagonal.

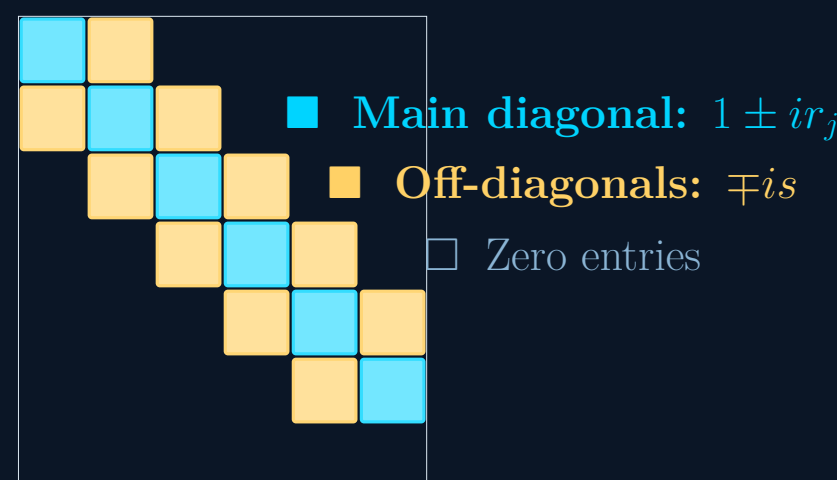
The Cayley Transform & Unitarity Proof	The CN propagator is the Cayley transform of H : $U_{CN} = A^{-1}B = \frac{I - i\gamma H}{I + i\gamma H}$ Since H is Hermitian: $A^\dagger = (I + i\gamma H)^\dagger = I - i\gamma H = B$. Therefore: $U_{CN}^\dagger U_{CN} = B^\dagger (A^\dagger)^{-1} A^{-1} B = A \cdot A^{-1} \cdot A^{-1} \cdot A = I$ $\Rightarrow \ \Psi^{n+1}\ ^2 = \ \Psi^n\ ^2 \quad (\text{exact at every time step})$ <i>CN is the unique 2-level scheme that is both $\mathcal{O}(\Delta t^2)$ and exactly unitary.</i>
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Order of accuracy: Taylor-expanding (??) gives local truncation error $\mathcal{O}(\Delta t^2, \Delta x^2)$ — matching both spatial and temporal discretisations.

5. Tridiagonal Matrix Structure of A and B

Let $r_j \equiv \gamma(2\beta + V_j)$ (diagonal) and $s \equiv \gamma\beta$ (off-diagonal). The linear system (??) reads:

$$\underbrace{\begin{pmatrix} 1+ir_1 & -is & 0 & \cdots & 0 \\ -is & 1+ir_2 & -is & \cdots & 0 \\ 0 & -is & 1+ir_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -is & 1+ir_{N-1} \end{pmatrix}}_A \Psi^{n+1} = \underbrace{\begin{pmatrix} 1-ir_1 & +is & 0 & \cdots & 0 \\ +is & 1-ir_2 & +is & \cdots & 0 \\ 0 & +is & 1-ir_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & +is & 1-ir_{N-1} \end{pmatrix}}_B \Psi^n$$



Sparsity pattern of A (identical structure for B ; signs flipped)

6. Thomas Algorithm — $\mathcal{O}(N)$ Tridiagonal Solver

Naïve LU decomposition costs $\mathcal{O}(N^3)$. The Thomas algorithm exploits the **tridiagonal structure** to achieve $\mathcal{O}(N)$ with only $\approx 8N$ floating-point operations — no fill-in.

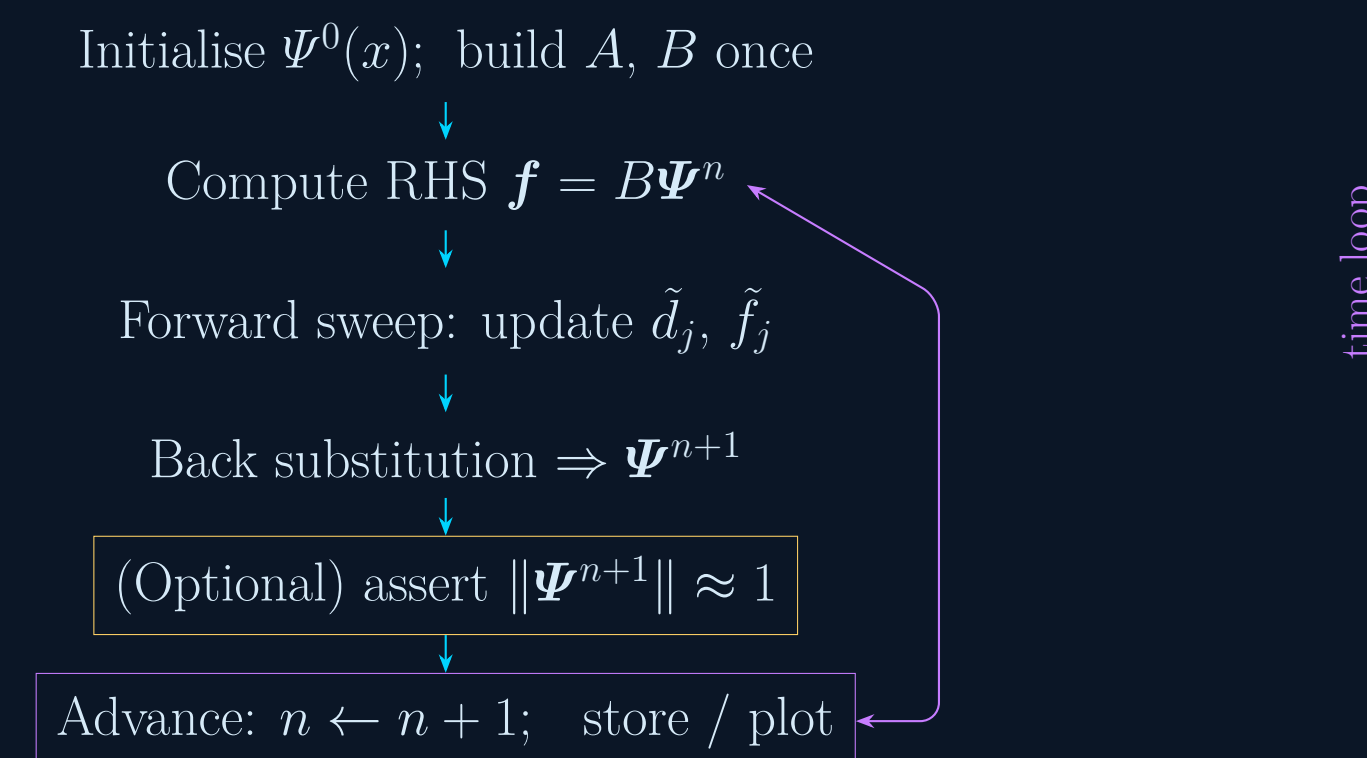
Let the system be $a_j \Psi_{j-1} + d_j \Psi_j + c_j \Psi_{j+1} = f_j$ with $a_j = c_j = -is$, $d_j = 1 + ir_j$.

Step 1 — Forward Sweep (modify coefficients)	Initialise: $\tilde{d}_1 = d_1, \tilde{f}_1 = f_1$. For $j = 2, 3, \dots, N-1$: $m_j = \frac{a_j}{d_{j-1}}, \quad \tilde{d}_j = d_j - m_j c_{j-1}, \quad \tilde{f}_j = f_j - m_j \tilde{f}_{j-1}$ Eliminates the sub-diagonal in $\mathcal{O}(N)$ steps.
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Back substitution starts from $j = N$ and proceeds backwards to $j = 1$.

Step 2 — Back Substitution (recover Ψ^{n+1})	$\Psi_{N-1}^{n+1} = \frac{\tilde{f}_{N-1}}{\tilde{d}_{N-1}}, \quad \Psi_j^{n+1} = \frac{\tilde{f}_j - c_j \Psi_{j+1}^{n+1}}{\tilde{d}_j}, \quad j = N-2, \dots, 1$
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Full time-stepping algorithm:



7. Method Comparison at a Glance

Scheme	Stability	Norm	Accuracy	Cost/step
Forward Euler (FTCS)	Unstable	Grows	$\mathcal{O}(\Delta t, \Delta x^2)$	$\mathcal{O}(N)$
Backward Euler (BTCS)	Stable	Decays	$\mathcal{O}(\Delta t, \Delta x^2)$	$\mathcal{O}(N)$
Crank–Nicolson	Stable	Exact	$\mathcal{O}(\Delta t^2, \Delta x^2)$	$\mathcal{O}(N)$
Dense LU solve	Stable	Exact	$\mathcal{O}(\Delta t^2, \Delta x^2)$	$\mathcal{O}(N^3)$

CN with Thomas algorithm delivers the best possible combination: $\mathcal{O}(N)$ cost, $\mathcal{O}(\Delta t^2)$ accuracy, and exact unitarity — simultaneously.

8. Summary & Physical Significance

The Crank–Nicolson method is a symplectic integrator, preserving the long-term energy structure of the system. It is the preferred method for simulating quantum wave packet dynamics.

Complete Crank–Nicolson Recipe	Grid: choose $\Delta x, \Delta t$; set $\beta = \hbar^2/(2m\Delta x^2)$, $\gamma = \Delta t/(2\hbar)$. Matrices: build tridiagonal $A = (1+i\gamma H_{FD})$ and $B = (1-i\gamma H_{FD})$ — <i>once</i>. Time loop: compute $f = B\Psi^n$; solve $A\Psi^{n+1} = f$ via Thomas Algorithm. Verify: $\ \Psi^{n+1}\ = 1$ to machine precision (guaranteed by U_{CN}).
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